

A32NX native weather radar on the ND

A native WASM gauge (ndwxr) drawing MSFS weather radar data on both NDs

Branch	feature/a32nx-nd-weather-radar
Aircraft	A32NX
Commits	5 commits, tip 5b8a3604d (PR 1)
Base	upstream master 2baa2b35e
Status	In develop. Flown by the user. PR draft 01.

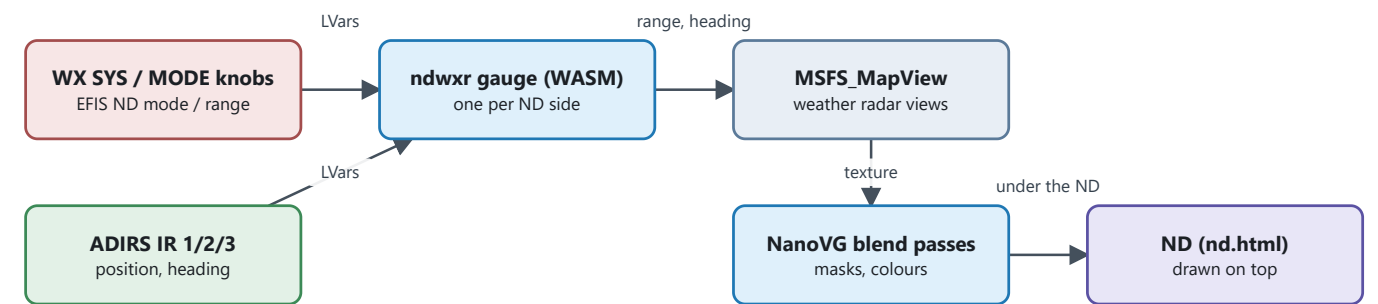
Purpose

The A32NX ND had no weather radar: the WX SYS / MODE knobs did nothing. The JS/Bing map radar path was found inconsistent between sessions; MSFS's native `MSFS_MapView` radar data was consistent, so the radar is a native gauge stacked on the ND.

Specification

- Drawn on the map pages of both NDs (ARC forward half disk, ROSE pages; the manuals pass extends it to ROSE ILS / VOR), aligned with the range rings; nothing on PLAN or with the radar off.
- WX SYS knob (SYS 1 / SYS 2 / OFF) and MODE knob: WX precipitation, WX+T precipitation + turbulence, TURB turbulence only, MAP a ground map (manuals pass).
- Colours by rain rate: green 0.7, yellow 4, red 12 mm/h; magenta turbulence proxy (rain rate > 20 mm/h, within 40 NM). Each threshold is a binary mask in a colour channel, blurred, saturated and squared: solid areas with crisp edges.
- Two independent NDs: CPT on AC ESS, F/O on AC 2; position from IR 1 / IR 2 or IR 3 per the ATT HDG knob.
- Build: igniter task `systems-ndwxr`, `wasm-opt` sign-extension lowering (as `terronnd`); `panel.cfg` gauges on `VCockpit02` / `VCockpit15`.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Limits of MSFS that shaped it

- No readback of the MapView texture (everything is blend passes); no turbulence data (proxy); no tilt or gain; at most 8 MapViews per module.

PR description (from the PR draft)

From `pr-drafts/01-wx-radar-a32nx.md`, section "Summary of Changes" onwards.

Summary of Changes

Adds the weather radar to the A32NX ND. It is drawn by a new native WASM gauge (`ndwxr`), stacked on the ND gauge of each side, from MSFS's own weather radar data (`MSFS_MapView`), so the ND JS code does not need the Bing map path.

- **Display:** ARC (forward half disk) and ROSE NAV (forward half), aligned with the ND's range rings, on the CPT and the F/O ND. Nothing is drawn on the other ND modes, or when the radar is off.
- **Controls:** the WX SYS knob (SYS 1 / SYS 2 / OFF) and the MODE knob: WX draws the precipitation, WX+T the precipitation and the turbulence, TURB only the turbulence, MAP nothing.
- **Colours:** airborne radar levels of rain rate: green 0.7, yellow 4, red 12 mm/h, magenta for turbulence. The engine's rain rate is noisy texel by texel, so each threshold is a binary mask (in a colour channel of the `MapView`) that is blurred, saturated and squared, which gives solid areas with crisp edges instead of grain.
- **Turbulence** is a proxy: MSFS gives no turbulence data, only rain rate. Rain rate above 20 mm/h is drawn magenta, out to 40 NM.
- **Two independent NDs:** the CPT ND is on the AC ESS bus, the F/O ND on AC 2. The position source follows the ADIRS rule of the ND: IR 1 for the CPT, IR 2 for the F/O, or IR 3 when the ATT HDG knob routes it to that side. ND mode and range come from `A32NX_EFIS_{L,R}_ND_MODE` / `_RANGE`. The two WX knobs are shared.
- **Ground:** like a real weather radar it does not transmit on the ground. Nothing is drawn while both main gear legs are compressed on either LGCIU. The engine already stabilises the beam in pitch and bank.
- **Build:** new igniter task `systems-ndwxr` (`build-a32nx:ndwxr`), the module is built with the same `wasm-opt` sign-extension lowering as `terrond`, `panel.cfg` gets the gauge on the two ND blocks (`VCockpit02` / `VCockpit15`, with the side as the last parameter), and `docs/README.md` lists the new folder.

Limits of MSFS that shaped it: no readback of the `MapView` texture (so everything is blend passes), no turbulence data, no tilt or gain, and the radar image is only drawn when it is airborne with weather around (use the MSFS weather presets to test).

Cockpit API Changes

No new LVars, no LVar changed. The gauge reads: `L:A32NX_EFIS_{L,R}_ND_MODE`, `L:A32NX_EFIS_{L,R}_ND_RANGE`, `L:XMLVAR_A320_WeatherRadar_Sys`, `L:XMLVAR_A320_WeatherRadar_Mode`, `L:A32NX_ADIRS_IR_{1,2,3}_LATITUDE` / `_LONGITUDE` / `_TRUE_HEADING`, `L:A32NX_ATT_HDG_SWITCHING_KNOB`, `L:A32NX_ELEC_AC_ESS_BUS_IS_POWERED`, `L:A32NX_ELEC_AC_2_BUS_IS_POWERED`, `L:A32NX_LGCIU_{1,2}_{LEFT,RIGHT}_GEAR_COMPRESSED`.

Screenshots (if necessary)

TO ADD (same camera, zoom and time of day, before/after): the CPT ND in ARC and in ROSE NAV with rain and a thunderstorm at 40 and 80 NM; WX and WX+T side by side; the F/O ND showing a different range; the ND on the ground (no radar). Before: the same ND with nothing drawn.

References

- MSFS SDK, `MSFS_MapView.h` native weather radar API (implementation, not real-life behaviour).
- **TO ADD** (the guide asks for real-life references): the A320 FCOM/PSCOM pages showing the weather radar on the ND (colour coding of precipitation and turbulence, WX / WX+T / TURB modes, WXR not transmitting on the ground), with screenshots.

Additional context

The native `MapView` weather data was found to be far more consistent across sessions than the JS/Bing map path, which is why the radar is a WASM gauge. `ndwxr` serves both ND gauges from one module (state per gauge context),

and each ND creates two MapViews (precipitation and turbulence). MSFS allows at most 8 MapViews per module, which matters for the TERR ON ND PR that reuses this module.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Start a flight in the air with weather (MSFS weather presets: rain, thunderstorm). Set the ND to ARC or ROSE NAV, WX SYS on SYS 1 and MODE on WX: precipitation appears in green/yellow/red, aligned with the range rings. Change the range (10 to 320) and switch ARC/ROSE NAV: the image stays aligned.
2. MODE on WX+T: magenta turbulence cells appear over strong cells (within 40 NM). TURB: only the magenta. MAP: nothing.
3. WX SYS on OFF, or the ND on PLAN / ROSE ILS / ROSE VOR: no radar.
4. On the ground (gear compressed): no radar. After take-off it comes back.
5. F/O ND: set another range and mode than the CPT ND: both show their own image. Switch off AC ESS (CPT) and AC 2 (F/O): the corresponding radar goes off. Try the ATT HDG knob (CAPT/F/O 3) on the position source.
6. Restart MSFS with the same weather and check the same cells appear (consistency between sessions).
7. Check the ND lines, range rings, traffic and flight plan stay visible over the weather.